



CALIBER 2026 - CASE 2: INTELLIGENT MANUFACTURING

One file. One truth. From data to decision.

380 incidents over 31 months, read from a single governed table. No server, no internet, no install.

\$67.19M

USD exposure in scope

2261 h

downtime, 380 records

179

overdue actions (109 past 1 year)

87.1%

AI retrieval at rank 1

MONTHLY LOSS EXPOSURE - JAN 2024 TO JUL 2026



Peak \$5.56M - average \$2.17MM per month. The full 31-month curve is on slide 2.
Jenny · Hafidz · Naufal - Advisor: Mr. Bayu Aji

The loss curve is going **up**, not down.

FIG. 1 – MONTHLY LOSS EXPOSURE, 31 MONTHS (KUSD)



Source: baseline Incident Database, 380 records x 23 columns. Red = Apr 2026 peak \$5.56M.
Dashed reading: 2026 run-rate \$2.64M/mo vs 2024 \$1.97M/mo (+34%).

2026 VS 2024

+34%

run-rate per month

WORST MONTH

\$5.56M

April 2026

BEHIND OVERDUE

56%

of exposure, ownerless

STALE > 1 YR

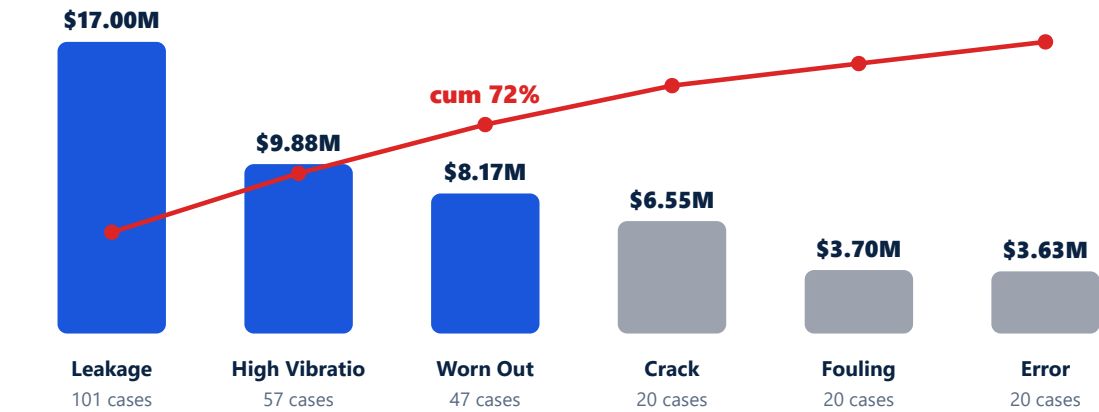
109

actions, oldest 926 d

- Six functions, six dashboards, six KPI definitions — hours validating before deciding.
- 220 actions open, 179 past due, median age 463 days.

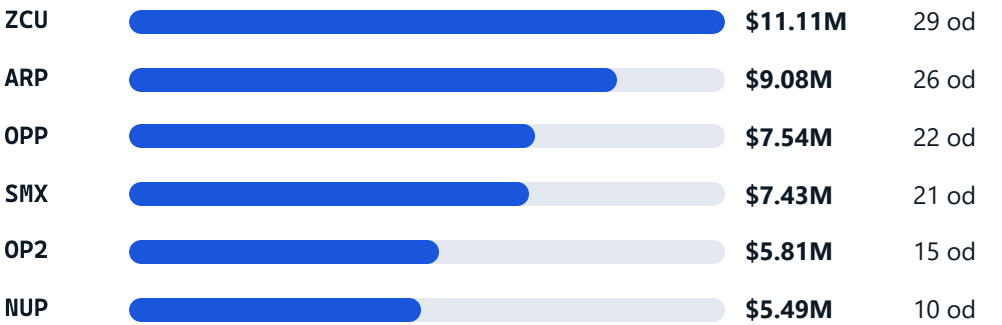
What we inherited — **measured**, not asserted.

FIG. 2 — EXPOSURE BY FAILURE MECHANISM (PARETO)



Source: incident mechanism column. Blue = top 3 (205 of 380 cases, \$35.04MM = 52% of exposure). Red dashes = cumulative share.

FIG. 3 — WHERE THE MONEY IS, BY UNIT



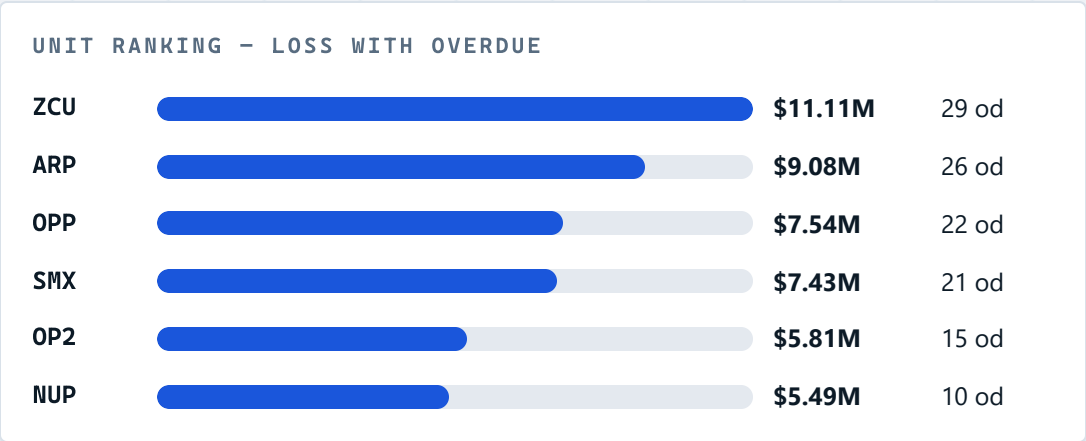
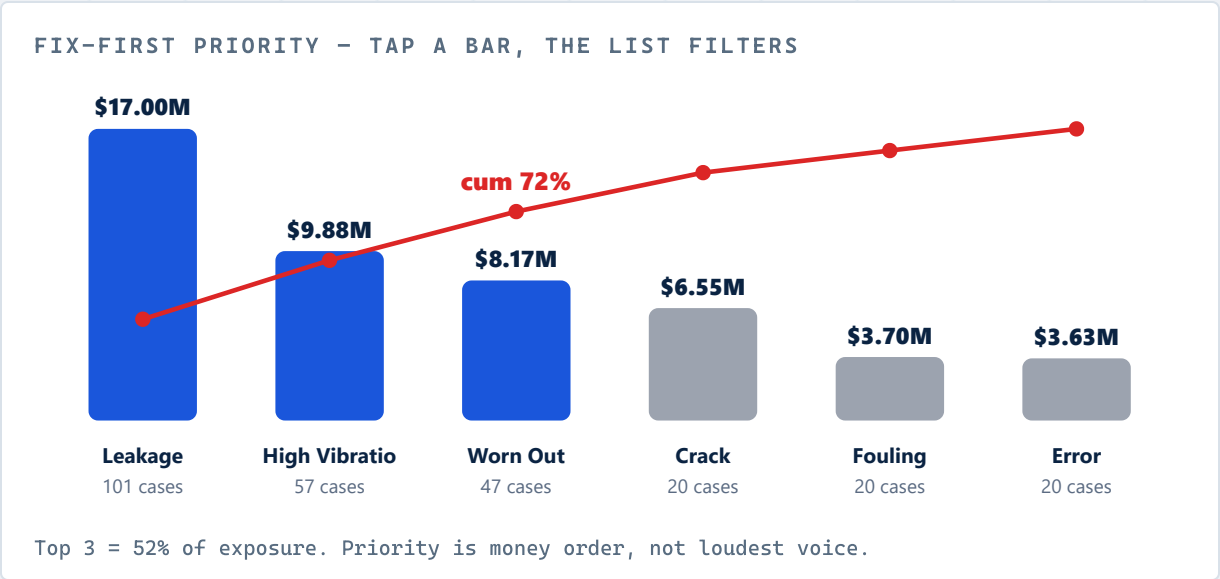
Bar = loss USD. od = overdue actions behind that unit. Unit ZCU alone: \$11.11MM + 29 overdue.

Four sources in. One governed table out.

Every chart on every view reads the same incident table. Edit one record, every figure recomputes.



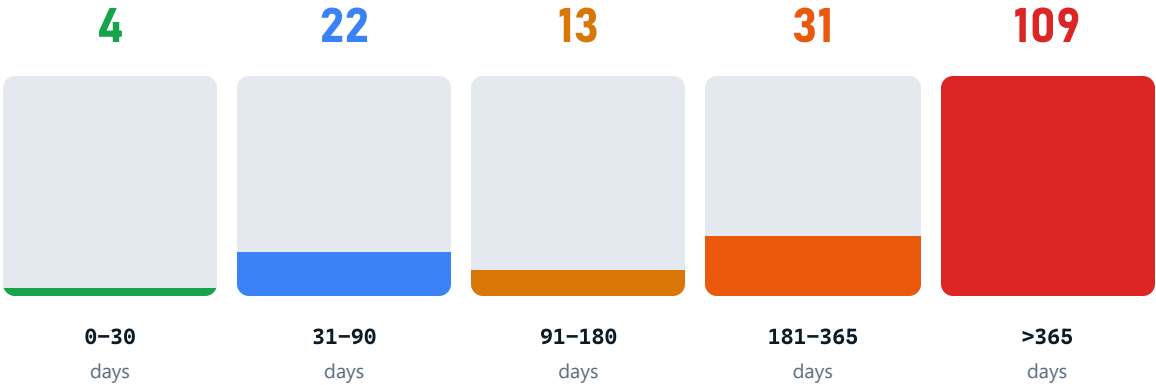
The executive view, in one screen.



- 8 KPI cards + trip logsheet: one row per trip, colored duration, follow-up dot.
- 0 charts without a data-provenance link.

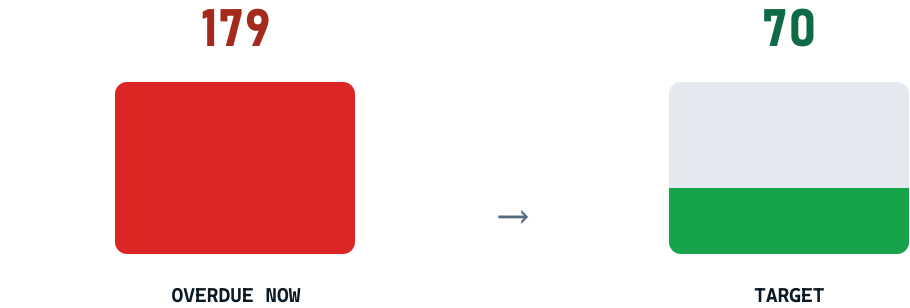
109 actions are more than a year late.

FIG. 4 – OVERDUE ACTIONS BY AGE, DAYS (179 TOTAL)



Measured at 19-Aug-2026. Median 463 d, oldest 926 d. \$37.32MM of exposure (56%) sits behind these rows.

BEFORE VS TARGET – MODELED COMMITMENT



109 actions >1 year old drop out

Basis: owner + due date mandatory at creation; closure blocked without evidence.
Not a measured result.

Red dot = nobody on it. Purple = in progress. Green = done.

AI suggests. Humans close.

SIMILAR-INCIDENT RETRIEVAL, BEFORE VS AFTER

69.2%



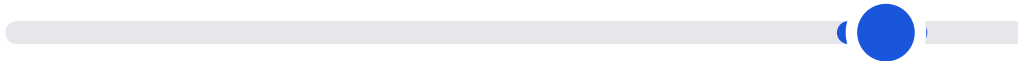
BASELINE MODEL

87.1%



FULL MODEL

+17.9 pts from uniting 4 sources



83.4%

90.1%

Both measured leave-one-out on all 380 records. 95% CI 83.4–90.1%. Method in Model Evaluation tab.

RANK 1

87.1%

full model

LIFT

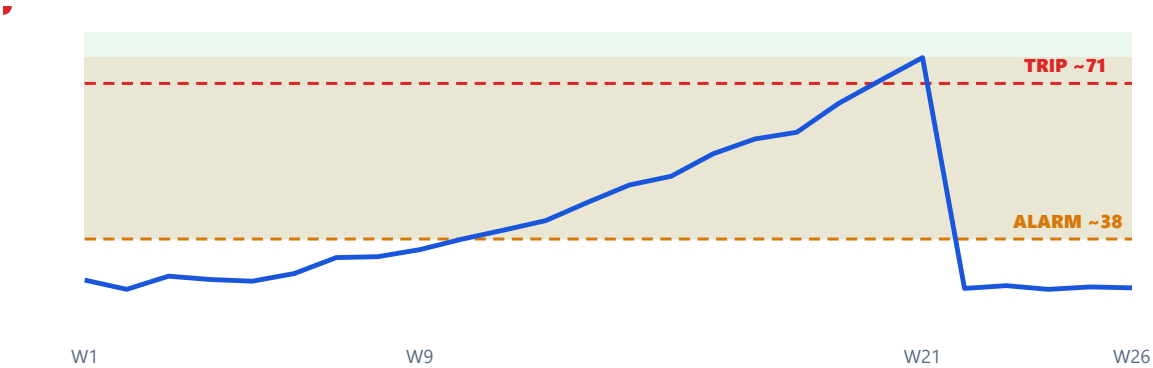
+17.9

pts vs thin baseline

- Tank > indication > prioritized alerts > tracked action with guidance.
- **Top-10** exposure plan in one click, owner + PM schedule each row.
- AI never closes an action. The PIC does.

Watch it degrade. 67 days of warning.

FIG. 5 - KO-3201 DE VIBRATION, 26 WEEKS (MICRON) - LIVE TRACE



Real condition history. ALARM week 10, TRIP week 21: 77 days to act. Live replay runs on screen.

Reading: 76.5 μm - TRIP · State: TRIPPED

WARNING WINDOW PER ASSET - FIRST ALARM TO TRIP

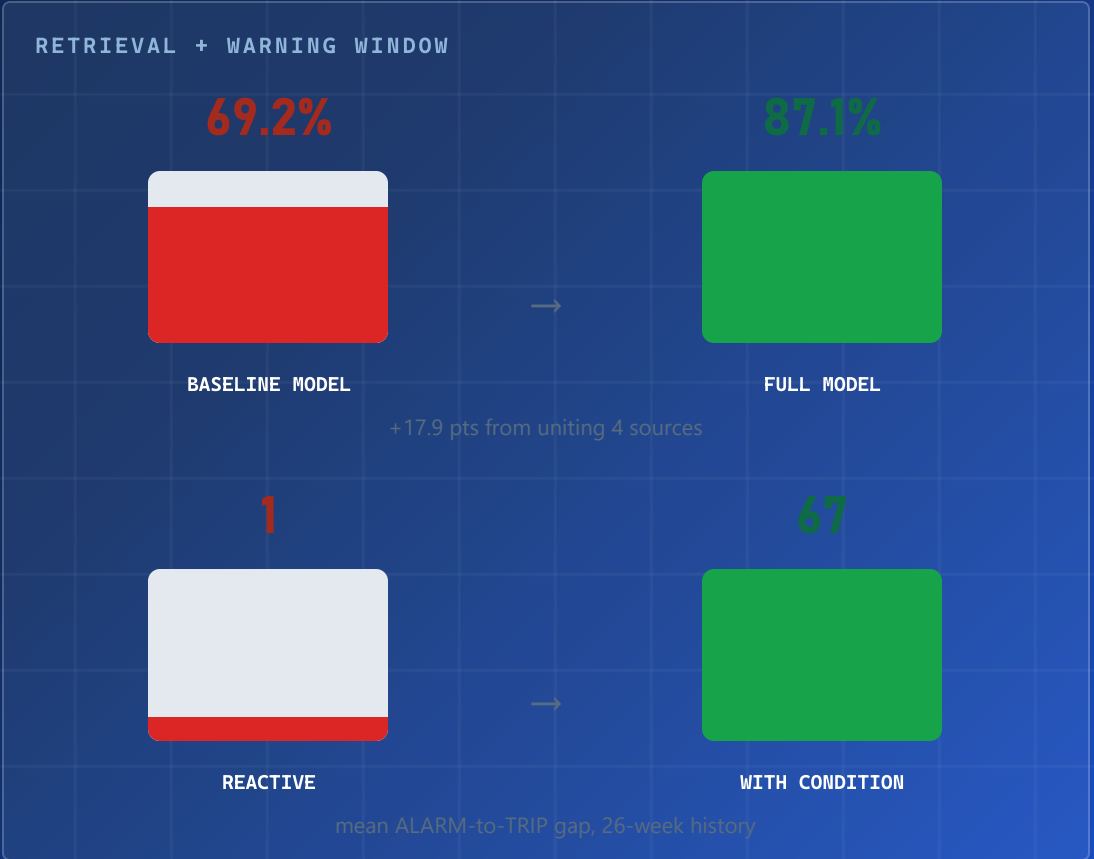


Mean 67 days across 5 assets. All 5 RCA decks integrated (4P / 4M+1E, CAPA, PIC, PM schedule).

■ Signal > alarm > trip > loss > owner + CAPA — one tap from any row.

Four numbers we commit to moving.

Baseline = measured. Target = modeled commitment with basis stated, in the open.



Measured. Labeled. Out of scope.

MEASURED

- Loss, downtime, aging — 380 real records
- Retrieval 87.1% / 98.4% leave-one-out
- Warning 67 d from history

LABELED

- Energy = derived proxy, formula printed
- CCTV = illustration, aggregates real
- Defects listed in Data Governance

OUT OF SCOPE

- No live streaming (not required)
- No auto-closure (human-in-loop)
- No extra data beyond 2 declared

TEAM & NEXT

Jenny · Hafidz · Naufal

Advisor: Mr. Bayu Aji. Built from the official baseline only, plus two declared additions.



DECK + NARRATIVE



VOICE + DEMO



DATA + BUILD

1 - METER IT

5 metered energy tags replace the proxy

2 - CONNECT IT

EDMS / AIMS / Twin for asset context

3 - STREAM IT

PI live feed, decoupled already

One file. One truth. From data to decision.

CALIBER 2026 - Case 2: Intelligent Manufacturing

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